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Via de driehoeksongelijkheid: $|x^2 - y^2| \leq |x^2| + |y^2| = x^2 + y^2$, dus:

$$0 \leq \left| \frac{x^2 - y^2}{\sqrt{x^2 + y^2}} \right| \leq \frac{x^2 + y^2}{\sqrt{x^2 + y^2}} = \sqrt{x^2 + y^2}$$

$$\lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2} = 0:$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{\sqrt{x^2 + y^2}} = 0$$

References